CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in machine learning on industrial and time-series data, from data pipelines and feature engineering to cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by ML work on process sensor data for batch prediction and on time-series forecasting.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full data and knowledge-extraction pipeline end-to-end, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction models with LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) on complex regulatory and technical documents, validated against a ground truth derived from SAP (ERP) data: precision/recall from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Delivered as an AI-augmented engineer, using Claude Code, custom skills and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, SQL, Postgres, MongoDB, Neo4j, LangGraph, LiteLLM, RAG, FastAPI, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application with LangGraph that routes each request across vector database queries, general LLM inference and summarization, after mapping business stakeholders' decision-making workflows.
- Developed a Python backend with FastAPI and a React front-end, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG generation and retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.

- **Tech stack: Python, LangChain, Hugging Face Transformers, PyTorch, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK), Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models on production process data to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Production Yield & Anomaly Pipeline | Airflow | dbt | XGBoost | MLflow | FastAPI

- Airflow + dbt pipelines from simulated MES and ERP data into Postgres; XGBoost yield prediction vs baseline and Isolation Forest for anomalies in finance postings, tracked and versioned in MLflow.
- FastAPI model serving in Docker with drift detection (PSI) that triggers a retraining workflow; documented features, model card and deployment configuration.

TECHNICAL SKILLSET

- **ML:** Scikit-Learn, PyTorch, TensorFlow/Keras, XGBoost, Random Forest, LSTM, time-series (ARIMA, SARIMA, Prophet), feature engineering
- **Data:** Python, SQL, Postgres, MongoDB, Neo4j, Pandas, NumPy, SciPy, Matplotlib, Streamlit
- **Deployment:** Docker, Kubernetes, FastAPI, AWS (CDK, Lambda, SQS, DynamoDB, EC2, S3), Azure, Git, GitLab CI/CD
- **GenAI:** LangGraph, LangChain, LiteLLM, RAG, Hugging Face Transformers, RAGAS evaluation
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT